



شبكة الجزيرة الإعلامية
ALJAZEERA MEDIA NETWORK

HTML5 LIVEBOX CONTROLLER

Using Node.js Middleware

PREPARED BY ALJAZEERA CREATIVE SOLUTIONS

1. Overview

HTML5 Livebox Controller is a browser-based control interface designed to simplify the operation of Livebox graphics within the existing Vizrt production environment.

Using **Node.js as middleware**, the controller integrates with **Media Sequencer (MSE)**, **Viz Director**, and **Viz Engine**, allowing operators to manage Livebox layouts from a single interface.

SIMPLIFIED OPERATION

Single interface for Livebox layout preparation and playout control.

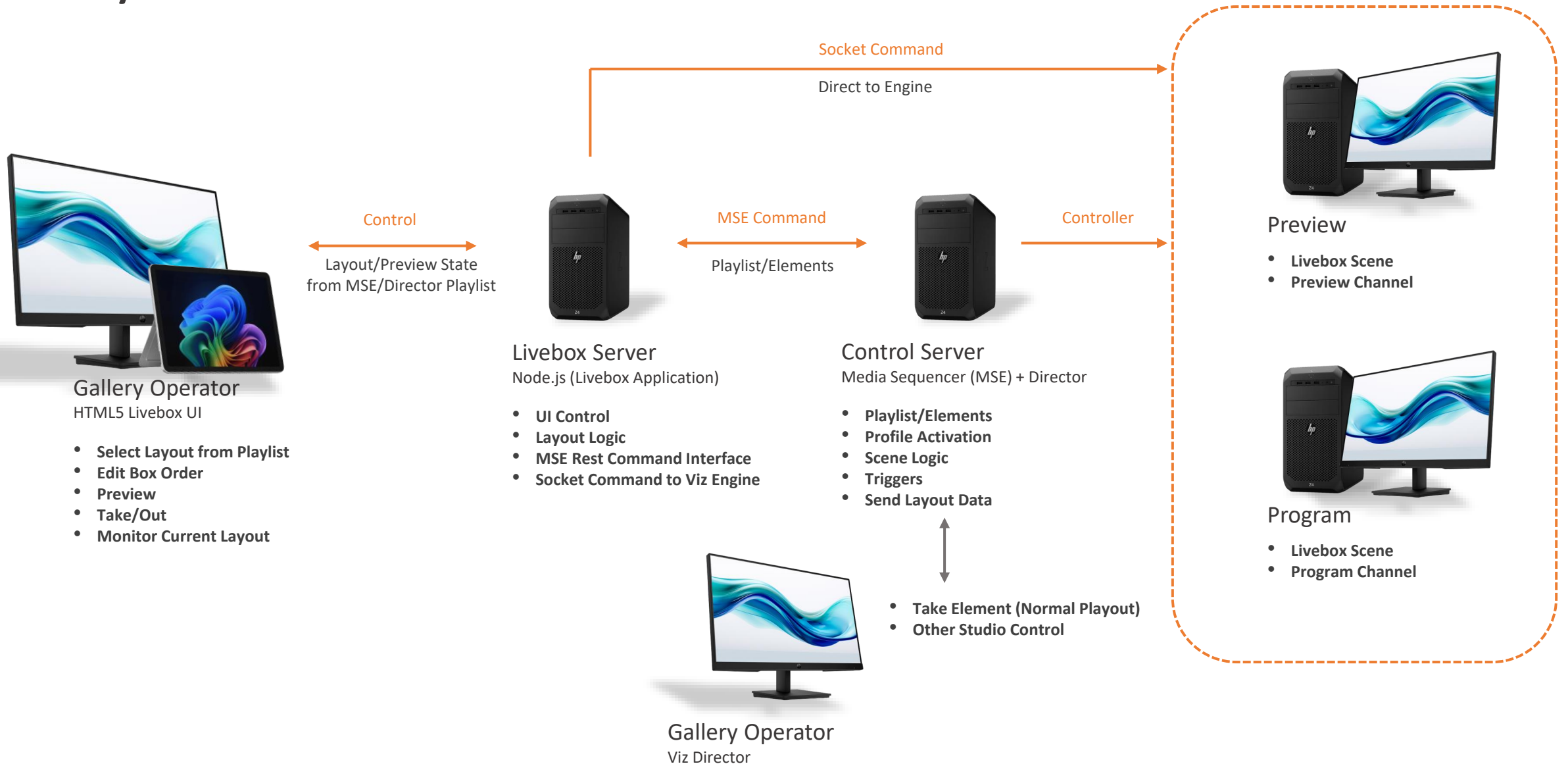
FLEXIBLE LAYOUT CONTROL

Reorder, remove, and manage Livebox based on production requirements.

VIZRT INTEGRATION

Integrates with the existing MSE, Viz Director, and Viz Engine environment through Node.js middleware.

2. System Architecture



3. System Requirements



Gallery Operator

HTML5 Livebox UI

NEW

Used to operate the Livebox, select layouts, edit box order, preview, and control playback.

Requirements

- Windows 10/11
- Google Chrome / Microsoft Edge
- Full HD display (1920 × 1080)
- Network access to Livebox Server

Notes

- Touchscreen Device.



Livebox Server

Node.js (Livebox Application)

NEW

Runs the HTML5 Livebox application, handles layout logic, and communicates with Media Sequencer and Viz Engines.

Hardware Specification

- HP Z4 G4 or equivalent
- Intel Core i5 / i7 or higher
- 16 GB RAM (minimum)
- 256 GB SSD or higher
- 1 GbE network

Software

- Windows 10/11 or Windows Server
- Node.js (LTS)
- HTML5 / JavaScript (Livebox UI)



Control Server

Media Sequencer (MSE) + Director

EXISTING

Handles playlists, element control, profile activation, and scene logic. Receives commands from Livebox and communicates with Viz Engines.

Software (Existing)

- Media Sequencer (MSE)
- Viz Director
- Existing broadcast configuration

Requirements

- Network access from Livebox Server
- No additional hardware required.



Preview

Preview

EXISTING

Renders the Livebox graphics for preview. Receives commands from Livebox and MSE/Director.

Software (Existing)

- Viz Engine
- Existing broadcast configuration
- Required Vizrt licenses

Requirements

- Network access from Livebox Server
- Socket command access from Livebox
- No additional hardware required.



Program

Program / On-Air

EXISTING

Renders the Livebox graphics for on-air output. Receives commands from Livebox and MSE/Director. **Viz Engine (Program).**

Software (Existing)

- Viz Engine
- Existing broadcast configuration
- Required Vizrt licenses

Requirements

- Network access from Livebox Server
- Socket command access from Livebox
- No additional hardware required.

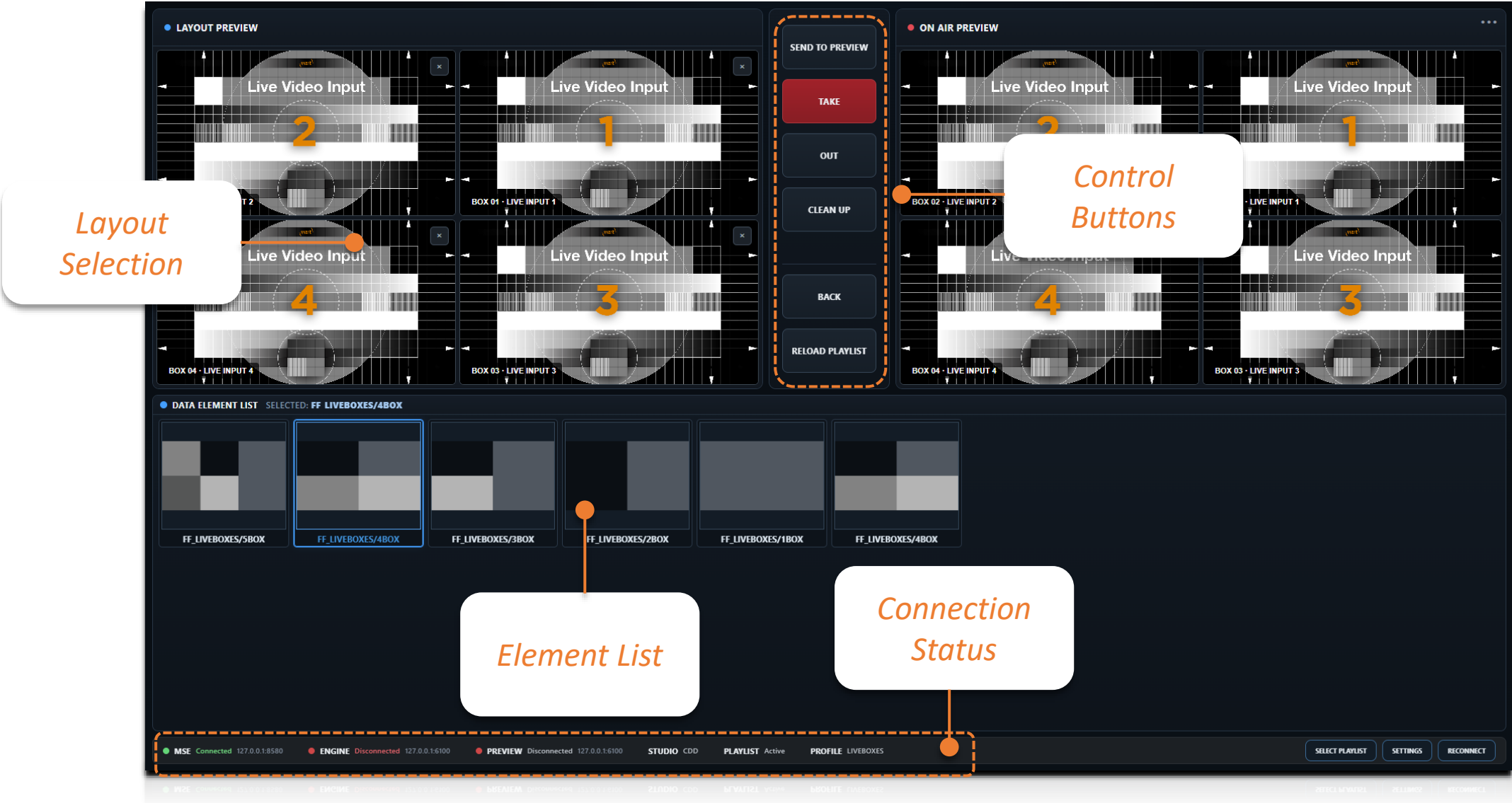
4. Network/Port Requirements

Source	Destination	Protocol	Port	Purpose
Gallery Operator	Livebox Server	HTTP	9091	Livebox Web UI / API
Livebox Server	Media Sequencer (MSE)	HTTP / REST	8580	Playlist, element & playout control
Livebox Server	Viz Engine Preview	TCP	6100	Direct Viz commands
Livebox Server	Viz Engine Program	TCP	6100	Direct Viz commands

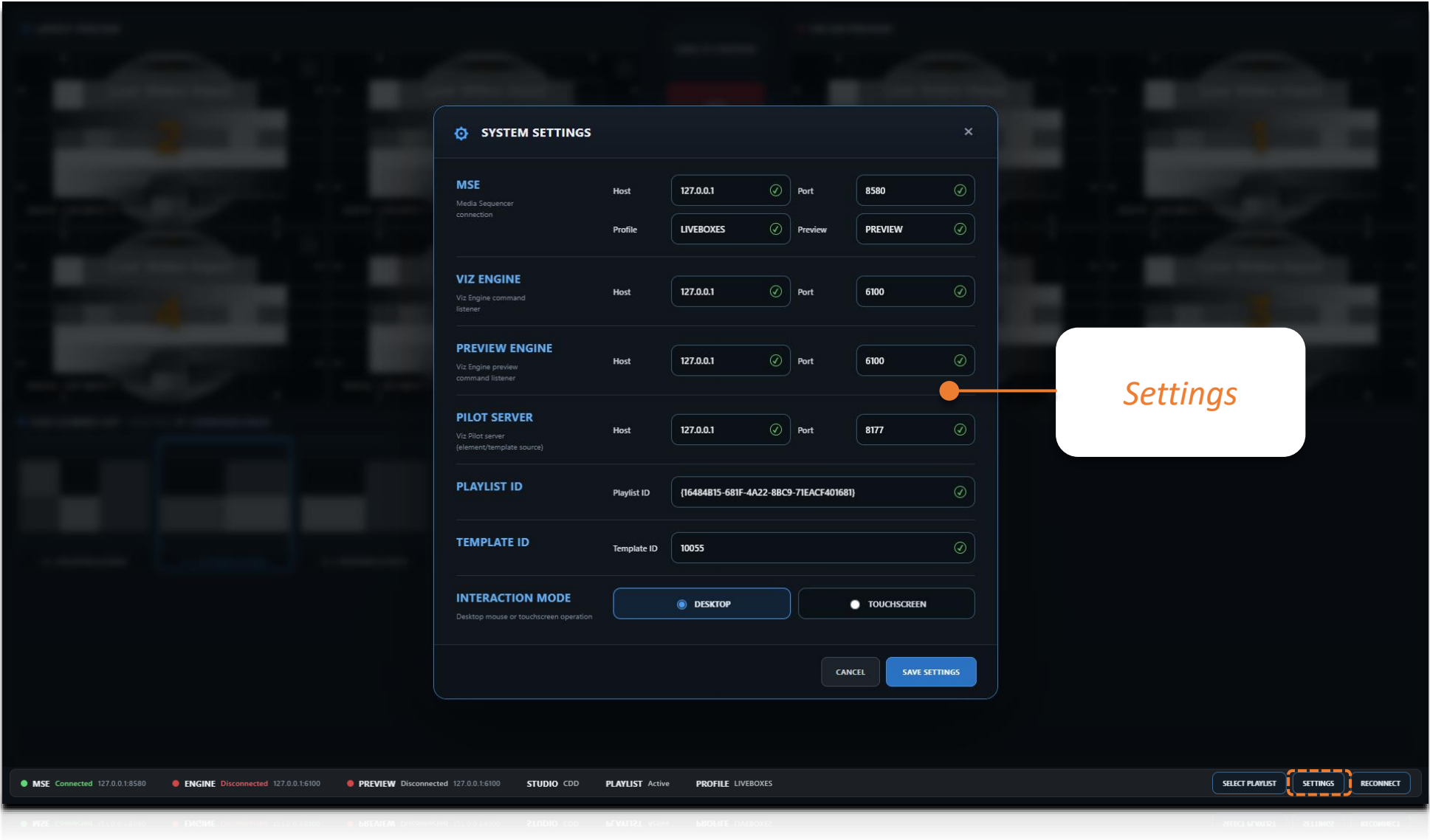
Network Requirements

- All required ports must be reachable between the respective systems.
- Firewall rules must allow the required inbound/outbound communication.
- All systems must be connected to the broadcast network.
- Static IP addresses or resolvable hostnames are recommended.

5. Operator Interface



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